

# Data Mining Complete Exam Notes

CT 72502 (Elective I) · BEI / BEX IV-I · TU, IOE · tiers over 23 papers, 2069–2082 BS

## 1. Introduction

2 hrs · in 22/23 papers · 2 topics, both TOP tier · no numericals in this ch.

Whole chapter in 2 items:

- KDD process **TOP 9/23**
- DW fundamentals **TOP 12/23**

### 1.1 Data Mining Origin

What is DM? + Explain steps of KDD process **2+3 2+5 2+6 1+5 4**

**TOP 9/23** 82 Bh, 79 Ba Most repeated Q in the paper.

#### DM

- Extraction of **interesting** patterns/knowledge from large data.
- Interesting = non-trivial + implicit + previously unknown + potentially useful.
- Not retrieval:
  - Query tool → answers what you already know to ask.
  - DM → finds patterns you didnt know to look for.
- Also called: KDD, knowledge extraction, data/pattern analysis, data archaeology, data dredging, BI.

#### Not DM

- Phone-book lookup, `SELECT AVG(salary)`, web search → info retrieval (query states exactly what is wanted).

#### Descriptive DM

- Characterise the general properties of data.
- Unsupervised.
- “What is in data? Unusual patterns?”
- Clustering, association, summarisation, visualisation.

#### Predictive DM

- Infer unknown  $Y = f(X)$  from known  $X$ .
- Supervised.
- “Which customers leave in 6 months?”
- Classification, regression, prediction.

#### DM vs KDD

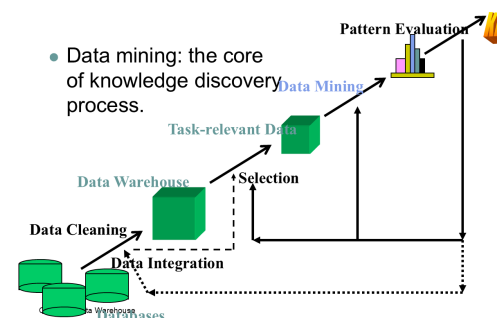
- Strictly: DM = **one step** of KDD (the step that runs the algorithms).
- In practice treated as synonyms. **IOE papers use them interchangeably.**
- **If Q says “steps of data mining” → write KDD steps.**

KDD: 7 steps (Han & Kamber) → default answer

1. **Data cleaning:** rm noise, smooth outliers, fill missing, fix inconsistency.
2. **Data integration:** combine multiple heterogeneous sources.
3. **Data selection:** retrieve only task-relevant data.
4. **Data transformation:** normalise, aggregate, generalise, discretise.
5. **Data mining:** apply algorithms. *Essential step.*
6. **Pattern evaluation:** keep truly interesting patterns (support, confidence, lift, novelty).
7. **Knowledge presentation:** visualise + represent for user.

Steps 1–4 = data preparation ≈ **60% of effort**.

### Data Mining: A KDD Process



**Conflict in your sources** **verified/added**

- Chapter 2\_Data Warehouse.ppt → **6 stages** (Adriaans & Zantinge): selection → cleaning → enrichment → coding → mining → reporting.
- Han & Kamber (syllabus ref. book 2) → **7 steps** above.
- Both examined at IOE (old CT 725 paper asked “Cleaning, Enrichment, Coding”).
- **Write 7-step version, add 1 line naming the 6-stage variant.**

### Answer skeleton [5–7]

1. DM definition, 1 line + “one step of KDD”.
2. **Draw the pipeline** (markers reward the fig).
3. 7 steps, **1 line each**. No more.
4. Close: “steps 1–4 = data prep ≈60% effort”.

## Origin of DM: why it emerged 2 3

### Confluence of disciplines

- DB systems
- Statistics
- Machine learning
- AI / pattern recognition
- Visualisation
- High-performance computing

### Drivers

- Automated collection → data explosion (POS scanners, txn logs, sensors, satellites, Web).
- Cheap storage → hoarding became rational. Analysis did not keep pace.
- Competition → unmined data = wasted asset.

### Why traditional/manual analysis fails

- Enormity (TB scale).
- **High dimensionality** (1000s of attrs).
- Sources heterogeneous + distributed.
- Data collected for **ops**, not analysis → noisy, incomplete, inconsistent.

### DM vs query tools (SQL)

- SQL: must know *exactly* what you want. DM: only *vaguely*.
- SQL → facts/aggregates. DM → patterns/trends.
- SQL: deterministic retrieval. DM: ML/statistical search.
- Hidden relations:
  - SQL cant trace them
  - DM finds them automatically

## “The world is data rich but information poor”. Justify. 8

PIN 1/23 73 Shr Justify-in-own-words → needs structure, not definitions.

1. **State paradox:** TBs stored, decisions still on intuition. Data ≠ info ≠ knowledge.
2. **Data rich:** POS logs, bank txns, telecom call records, remote sensing, Web. Growth automated.
3. **Info poor, why:**
  - Volume defeats manual inspection.
  - Data is operational → noisy, incomplete, inconsistent.
  - Scattered across heterogeneous systems.
  - Query tools answer only what is asked.
  - Shortage of analysts.
4. **Consequence:** “data tombs” = archives written, never read. Decisions made w/o evidence already paid for.
5. **Resolution:**
  - DW consolidates + cleans
  - DM extracts the patterns
  - Data → Info → Knowledge → Decision
6. **Eg:** retail chain holds millions of receipts (rich) but cant say which products sell together (poor) until association analysis gives {nappies} ⇒ {beer}.

## “Data mining turns a large collection of data into knowledge”. Justify 6

81 Bh **The mirror image of the question above**, and it is asked separately, so learn it as its own answer. That one asks why we are stuck; this one asks how DM gets us out. **Structure the answer on the four-rung ladder** and carry one example all the way up it. verified/added

### The ladder, one rung at a time

1. **Data**: raw recorded facts, no context. Eg 2 million supermarket receipt lines.
2. **Information**: data aggregated and given context by querying. Eg “nappies sold 4,200 units last month”. A DW and OLAP get you here, and **this is as far as a query tool can go** — it answers only what you thought to ask.
3. **Knowledge**: patterns that hold across the data and **were not asked for**. Eg “customers buying nappies on Friday evening also buy beer”. Only **data mining** produces this rung, because only it searches the space of possible patterns.
4. **Decision / wisdom**: acting on the pattern. Eg move the beer display beside the nappies, and measure the lift in basket value.

### Why the step to knowledge is data mining’s alone

- **Queries are deductive, mining is inductive**. SQL retrieves what you specify; mining induces a general rule from the instances.
- It is **automatic and exhaustive**: Apriori tests every itemset the thresholds allow, far more hypotheses than an analyst would think to write.
- The output is **predictive**, not just descriptive: a classifier built on past data labels tuples that do not exist yet.
- The patterns are required to be **valid, novel, useful and understandable** — Fayyad’s four conditions. **“Novel” is what makes it knowledge**: a pattern you already knew is information, not knowledge.

### Close with the qualifier

- DM does not turn *all* data into knowledge. Garbage in, garbage out: without preprocessing (Ch 2) it turns noise into confident nonsense, and without pattern evaluation (§4.1) it reports rules that are strong but false.

## Architecture of a typical DM system (+Fig) 2 4+2

78 Bh asked as “Explain DM process with diagram”.

- **Data sources**: DB, DW, WWW, other repos.
- **DB/DW server**: fetches data relevant to the request.
- **Knowledge base**: domain knowledge guiding search + judging interestingness (concept hierarchies, user beliefs, constraints).
- **DM engine**: characterisation, association, classification, clustering, outlier, evolution analysis.
- **Pattern evaluation**: applies interestingness measures. *Best pushed into the engine* so dull patterns are never generated.
- **UI**: specify task, browse schemas/hierarchies, visualise results.

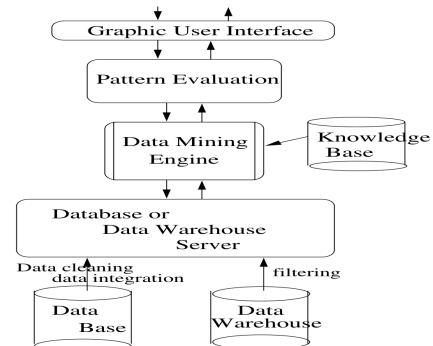


Figure 1.5: Architecture of a typical data mining system.

## DM functionalities 2 4

- **Characterisation & discrimination**:
  - summarise the target class
  - contrast it against a comparative class
- **Association & correlation**: freq patterns, rules  $X \Rightarrow Y$ . → Ch 4
- **Classification & prediction**: assign class labels / predict numeric. → Ch 3
- **Cluster analysis**: group w/o pre-defined labels. → Ch 5
- **Outlier / anomaly**: objects not complying w/ general behaviour. Noise to discard, but *the prize* in fraud. → Ch 6
- **Evolution analysis**: trends for objects changing over time. → Ch 7

## What is DM? Explain its applications. 10

PIN 1/23 71 Ch

| Domain                 | What is mined                                                                                                                                                                         |        |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| Biomedical / DNA       | Similarity search across DNA seqs; <b>association analysis</b> for co-occurring genes (most diseases = gene combinations, not 1 gene); path analysis linking genes to disease stages. |        |
| Financial              | DW design for multidim analysis; loan payment prediction, credit policy; classification + clustering for customer targeting; <b>money-laundering detection</b> .                      |        |
| Retail / market basket | Which products sell together; customer retention; campaign effectiveness; recommendation.                                                                                             |        |
| Telecom                | Fraudulent usage patterns; multidim analysis of calls; mobile user behaviour.                                                                                                         |        |
| Intrusion detection    | Anomaly detection over traffic + audit logs.                                                                                                                                          | → Ch 6 |
| Web mining             | PageRank, usage + structure mining.                                                                                                                                                   | → Ch 7 |

### 1.2 Data Mining & Data Warehousing Basics

## What is a DW? Write its key features. 2 2+4 2+5

TOP 12/23 73 Ch, 70 Asa, 69 Ch

### Definition (Inmon)

- DW = **subject-oriented, integrated, time-variant, non-volatile** collection of data supporting management decision-making.
- **Memorise as S-I-T-N** → this is the whole “key features” answer.
- **S = Subject-oriented**
  - Organised around subjects (customer, product, sales), not daily ops.
  - Data useless for decisions is excluded.
- **I = Integrated**
  - Built from heterogeneous sources (relational DBs, flat files, txn records).
  - Cleaning + integration force consistent naming, encoding, units.
  - → 1 name per entity.
- **T = Time-variant**
  - Long history (5–10 yrs).
  - Every key structure carries a time element.
  - Operational DB holds only the *current* value.
- **N = Non-volatile**
  - Physically separate. No operational update.
  - Only 2 ops: **initial load** + **access**.
  - → no txn processing, recovery or concurrency control needed.
- **Data warehousing** = the *process* of building + using a DW (extract, transform operational data, load into central store).

## How is DW diff from a DB? How are they similar? 2+2

PIN 3/23 75 Ash, 73 Ch Asks both directions. 73 Ch asks it as DW vs RDBMS.

|               | Operational DB (OLTP)               | Data warehouse (OLAP)                     |
|---------------|-------------------------------------|-------------------------------------------|
| Purpose       | Run day-to-day business             | Support analysis + decisions              |
| Content       | Current, detailed                   | Historical, summarised + detailed         |
| Design        | Highly normalised, many tables (ER) | De-normalised, few tables; star/snowflake |
| Access        | Short read/write txns               | Long, complex, read-mostly                |
| Users         | Clerks, DBAs, front-line            | Managers, analysts, execs                 |
| Unit of work  | Simple txn                          | Complex query                             |
| Records/query | Tens                                | Millions                                  |
| Size          | 100 MB–GB                           | 100 GB–TB                                 |

**Similarities** verified/added

- Both persistent, managed data stores.
- Both usually on a relational DBMS.
- Both use schemas, indexes, SQL.
- Both enforce integrity + security.
- DW is *built from* the DBs.

**Why keep them separate**

- **Performance:** opposite tuning.
  - DBMS → OLTP (indexing, concurrency, recovery)
  - DW → OLAP (multidim views, consolidation)
  - both on 1 system degrades both
- **Missing data:** decisions need history ops DBs dont keep.
- **Consolidation:** needs aggregation across heterogeneous sources.
- **Data quality:** inconsistent codes/formats reconciled once, centrally.

**Explain DW architecture with its analytical processing** 8

**TOP 9/23** 75 Ch 2 architectures in your sources: complementary, not contradictory (three-tier = layered, process = component-wise).

**(a) Three-tier architecture (Han & Kamber) → standard answer**

- **Bottom tier** = warehouse DB server (relational).
  - Data pulled via **gateways**: ODBC, OLE-DB, JDBC.
  - Back-end tools: extract, clean, transform, load, refresh.
  - Holds **metadata repo**.
- **Middle tier** = **OLAP server** (presents data as a cube).
  - **ROLAP:**
    - o extended relational DBMS
    - o maps multidim ops → relational ops
    - o scales to large data
  - **MOLAP:**
    - o special-purpose server, direct multidim arrays
    - o fast, but sparse data wastes space
  - **HOLAP:**
    - o hybrid of the two
    - o detail in relational, aggregates in arrays
- **Top tier** = front-end client: query & reporting, analysis, DM tools.

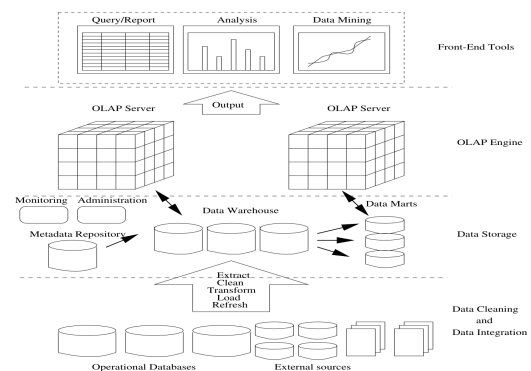
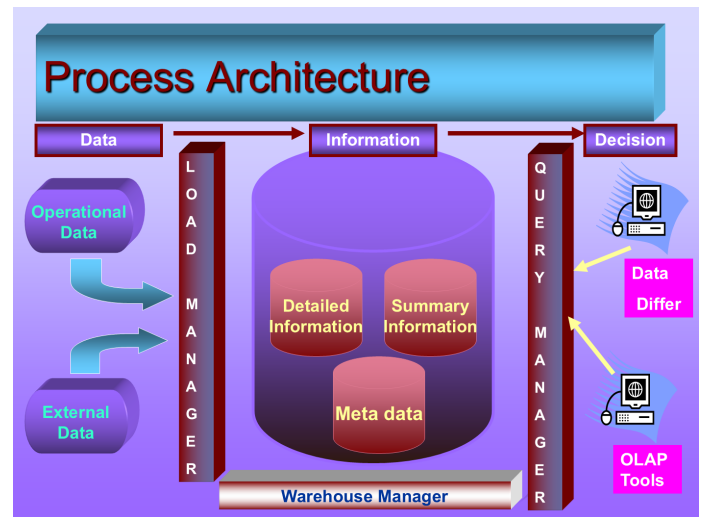


Figure 2.12: A three-tier data warehousing architecture.

**(b) Process / component architecture (form drawn in class notes)**

- **Load manager:** all extract + load ops. Fast-loads into temp store, simple transforms.
- **Warehouse manager:**
  - consistency + referential checks
  - merges temp store into the published DW
  - creates indexes + business views
  - updates aggregations
  - backup + archive
- **Query manager:**
  - directs queries to the right tables
  - stops 1 query starving the system
  - captures query profiles
- **Detailed info:** lowest-granularity data retained.
- **Summary info:**
  - pre-computed aggregations built by the warehouse manager
  - transient, changes as query profiles change
- **Metadata:** data about data. Describes how info is structured, maps sources to a common view.



### (c) "...with its analytical processing" → 3 kinds of DW application

1. **Information processing:** querying, basic stats, reporting (crosstabs, tables, charts). *No new knowledge.*
2. **Analytical processing (OLAP):** multidim analysis. Roll-up, drill-down, slice, dice, pivot. *Summarises, doesn't learn.*
3. **Data mining:** knowledge discovery from hidden patterns. *Creates new knowledge.*

### What is DW and data mart? (short notes) 15 2+4

PIN 3/23 70 Ch, 71 Shr, 69 Ch

#### Data mart

- Subset of a DW, confined to 1 user group / functional area, in its own DB.
- **Dependent:** sourced *from* an enterprise DW → consistent by construction.
- **Independent:** populated *directly* from sources, no central DW → faster to build, risks inconsistency between marts.
- **Benefit:**
  - less volume to scan → faster queries
  - each department owns its own structures
- **Eg:** departmental store sales mart, bank loan mart.

#### 3 DW models

- **Enterprise warehouse:** all subjects, whole org. Cross-functional, large, slow to build.
- **Data mart:** subset for a specific group.
- **Virtual warehouse:**
  - set of views over operational DBs
  - only some summary views materialised
  - easy to build, but loads the operational systems

## Describe Snowflake schema with example 4

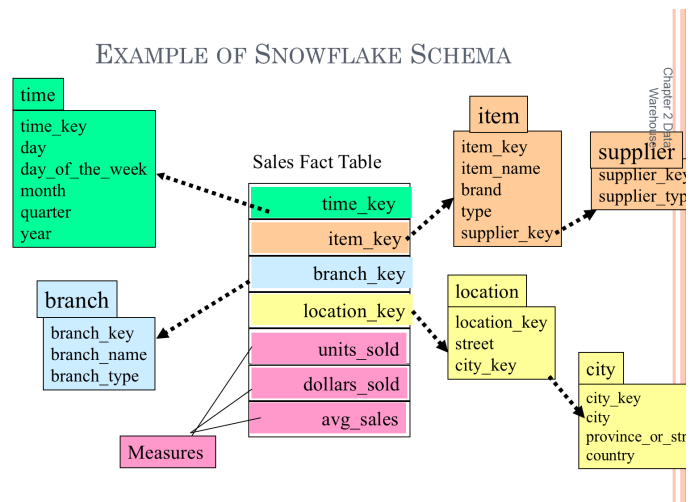
PIN 1/23 74 Ch

- **Star**: 1 central **fact table** (measures + FKs) + **de-normalised** dimension tables. Simple, fast joins.
- **Snowflake**: star refined, some dimension hierarchies **normalised** into smaller sub-dimension tables. Saves space, less redundancy, but more joins → slower.
- **Fact constellation (galaxy)**: multiple fact tables sharing dimension tables = collection of stars.

Eg: sales DW

- Fact Sales: time\_key, item\_key, branch\_key, location\_key, units\_sold, dollars\_sold, avg\_sales.
- location normalised: location → city → province.
- item → supplier.
- **That normalisation of the dimension hierarchy is what makes it snowflake, not star.**

→ Full multidim model (cuboids, base/apex, star-schema *drawing* numerical) sits in **Ch 2, OLAP & Multidimensional Data Analysis**, where it is examined harder.



## Fundamental differences between DM and DW 3

PIN 2/23 76 Ash

|          | Data warehousing                                                                                               | Data mining                             |
|----------|----------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| Nature   | Storage architecture                                                                                           | Analysis process                        |
| Purpose  | Consolidate + organise data                                                                                    | Discover hidden patterns in it          |
| Output   | Consistent query-able historical data                                                                          | Rules, models, clusters, predictions    |
| Done by  | Engineers / DBAs                                                                                               | Analysts, statisticians, business users |
| Relation | DW = infrastructure; DM = 1 of its 3 uses. DW not required for DM, but a clean DW makes DM far more effective. |                                         |

## How can a DM system be integrated with a DB/DW system? 4

PIN 1/23 78 Bh Answer = the 4 coupling schemes.

### 1. No coupling

- Uses no DB/DW function at all.
- Reads flat file → mines → writes to another file.
- Simple, but: poor data selection, no scalability, no consistency.

### 2. Loose coupling

- Mining system → fetches data from repo managed by DB/DW.
- Stores results back into a file / designated place in DB.
- Uses DBMS for storage + retrieval only.
- Cant exploit indexing or query optimisation → memory-bound, doesnt scale well.

### 3. Semi-tight coupling

- Besides linking, DB/DW provides efficient **mining primitives**.
- Primitives: sorting, indexing, aggregation, histogram analysis, multi-way join.
- + pre-computed stats: sum, count, max, min.
- → materially better performance.

### 4. Tight coupling

- DM system **smoothly integrated** into DB/DW as 1 functional component.
- Mining queries optimised alongside DB queries.
- Intermediate results can be stored.
- Scales. **Most desirable design.**

### 1.3 PYQ Mapping

Tiers: **TOP /23** = 9+/23 papers **HOT /23** = 5-8 **PIN /23** = 1-4 but  $\geq 8$  marks. Same Q in diff wording grouped under 1 entry.

#### 1.1 Data Mining Origin

- TOP 7/23** Define DM + explain KDD / DM process. 2+3 to 2+6
  - "What is data mining? Explain the process of data mining." 74 Ash
  - "...Explain all the steps of knowledge discovery." 72 Ch
  - "...What are the steps involved in knowledge discovery process?" 79 Bh
  - "...Explain the steps of KDD process briefly." 81 Ba
  - "...Explain general steps in brief." 72 Ka
  - "What is a Data Mining? Explain its application." 71 Ch → 2nd half swaps to applications.
- PIN 1/23** "The world is data rich but information poor". Justify. 8 73 Shr

#### 1.2 DM & DW basics

- HOT 3/23** What is a DW / key features + architecture.
  - "What is data warehousing? Explain with example where data warehouses are used." 2+4 80 Ba
  - "Write key features of data warehouse. Explain each step of the KDD process w/ example." 2+5 80 Bh
  - "Explain Data Warehouse architecture with its analytical processing." 8 75 Ch
- PIN 2/23** DM vs DW / DW vs DB.
  - "What are the fundamental differences between Data Mining and Data Warehousing? Describe the steps of KDD." 3+7 76 Ash
  - "How is data warehouse different from a database? How are they similar?" 2+2 75 Ash
- PIN 1/23** "Explain how DM system can be integrated with DB/DW system. Explain DM process with diagram." 4+2 78 Bh
- PIN 1/23** "What is data warehouse and data mart? Describe Snowflake scheme with example." 2+4 74 Ch
- PIN 2/23** Short notes: Data warehouse and Data mart. 15 70 Ch, 71 Shr

#### Papers added by the 2026-09 merge

The six papers of DM4.1BCT.pdf (2082 Bh, 2081 Bh, 2079 Ba, 2073 Ch, 2070 Asa, 2069 Ch), indexed here for the first time — the notes were built before that scan was merged. **N** = solved in this chapter's companion. verified/added

- What is data mining? Describe the steps involved in it. 2+6 82 Bh
- "Data mining turns a large collection of data into knowledge". Justify. 6 81 Bh verified/added
- KDD process + **the challenges that motivated data mining.** 3+3 79 Ba
- How is a data warehouse different from an RDBMS? Also list the similarities. 2+2 73 Ch
- Short notes: **Data warehouse** 70 Asa · **Datamart** 69 Ch. 5 5

#### Coverage check

- All Ch 1 PYQs covered above.
- Filled by research, absent from your notes verified/added :
  - Similarity half of "how are DW and DB similar?" 75 Ash (notes only contrast them).
  - Reconciliation of the **6-stage vs 7-step KDD** clash between your 2 sources.

#### Sources used for this chapter

- **Han & Kamber 3rd ed** (full textbook PDF): ch 1 KDD process, DM system architecture, functionalities, applications; ch 4 DW definition, 3-tier architecture, DW models, DW usage. **Primary authority; all of the above re-verified against it.**
- **Han & Kamber 1st ed manuscript** (ch01.pdf, ch2.pdf): source of the two architecture figures, and of the **4 coupling schemes**, which the 3rd ed drops. **78 Bh still examines coupling**, so it is kept here.
- Chapter 2\_Data Warehouse.ppt, Chapter 4 and 5 DW Architecture.ppt, applicationDM.ppt, data mining\_/Chapter 1\_2.pdf.
- **No DBK lecture exists for Unit-1**, so class slides could not be used here.

## Numericals

- **Ch 1 has no numerical component.** No Q in 23 papers needs a calculation here → no “Ch1 Numerical” companion. First numericals appear in Ch 2 (normalisation, similarity measures, PCA).

### Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

| Short    | Means                            | Short   | Means                         |
|----------|----------------------------------|---------|-------------------------------|
| DM       | data mining                      | DW      | data warehouse                |
| DB       | database                         | DBMS    | database management system    |
| KDD      | knowledge discovery in databases | DSS     | decision support system       |
| OLTP     | on-line transaction processing   | OLAP    | on-line analytical processing |
| ROLAP    | relational OLAP                  | MOLAP   | multidimensional OLAP         |
| HOLAP    | hybrid OLAP                      | ETL     | extract, transform, load      |
| BI       | business intelligence            | ER      | entity–relationship           |
| attr(s)  | attribute(s)                     | dim(s)  | dimension(s)                  |
| multidim | multidimensional                 | txn(s)  | transaction(s)                |
| info     | information                      | ops     | operations                    |
| repo     | repository                       | freq    | frequent / frequency          |
| config   | configuration                    | no.     | number                        |
| PCA      | principal component analysis     | SVD     | singular value decomposition  |
| SMC      | simple matching coefficient      | KNN     | $k$ -nearest neighbour        |
| ANN      | artificial neural network        | POS     | point of sale                 |
| FK       | foreign key                      | UI      | user interface                |
| GIGO     | garbage in, garbage out          | w/, w/o | with, without                 |
| rm       | remove                           | vs      | versus                        |
| Eg       | example / for example            | +Fig    | draw a figure                 |
| →        | leads to / therefore             | +       | and                           |

Exam-paper month codes (used in the year tags): Ba = Baishakh · Jth = Jestha · Asa = Ashar · Shr = Shrawan · Bh = Bhadra · Ash = Ashwin · Ka = Kartik · Mng = Mangsir · Po = Poush · Ma = Magh · Ch = Chaitra.

Tier chips: **TOP /23** = topic asked in 7 or more of the 17 papers · **HOT /23** = 4–6 papers · **PIN /23** = 1–3 papers but worth  $\geq 8$  marks. Mark chips such as **2+5** are the marks that question actually carried. **verified/added** marks anything researched and added that was not in the source notes.